

EC800G-CN QuecOpen Reference Design

LTE Standard Module Series

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About the Document

Revision History

Version	Date	Author	Description
-	2023-06-21	Jayce JIANG/ Frank WANG	Creation of the document
1.0	2023-08-16	Phoebe FU/ Frank WANG	First official release

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1 Reference Design

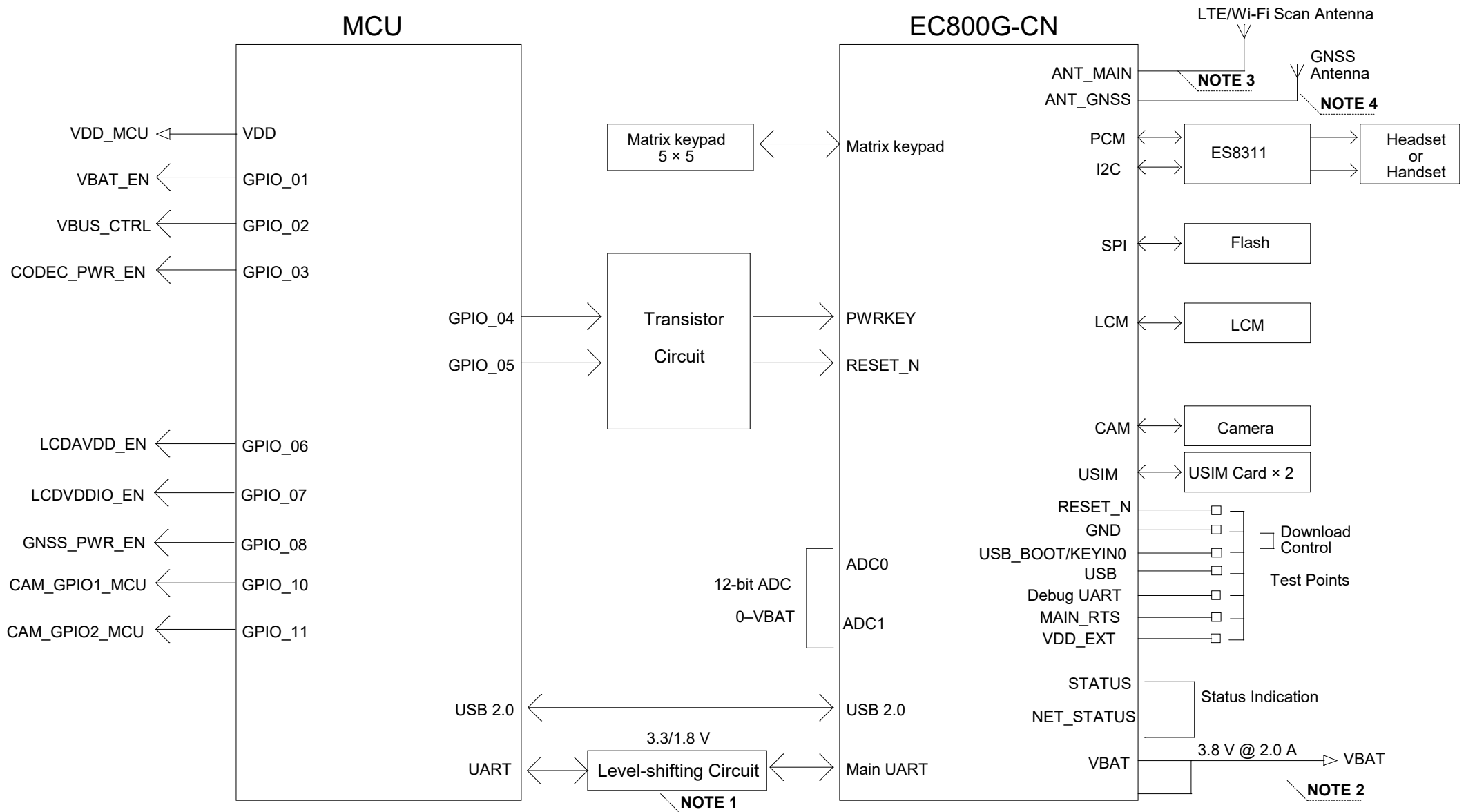
1.1. Introduction

This document provides the reference design for Quectel EC800G-CN QuecOpen® module. The reference design mainly includes block diagrams of power supply, antenna, USIM, UART, audio, LCM, camera, matrix keypad and external flash interfaces.

1.2. Schematics

The schematics illustrated in the following pages are provided for your reference only.

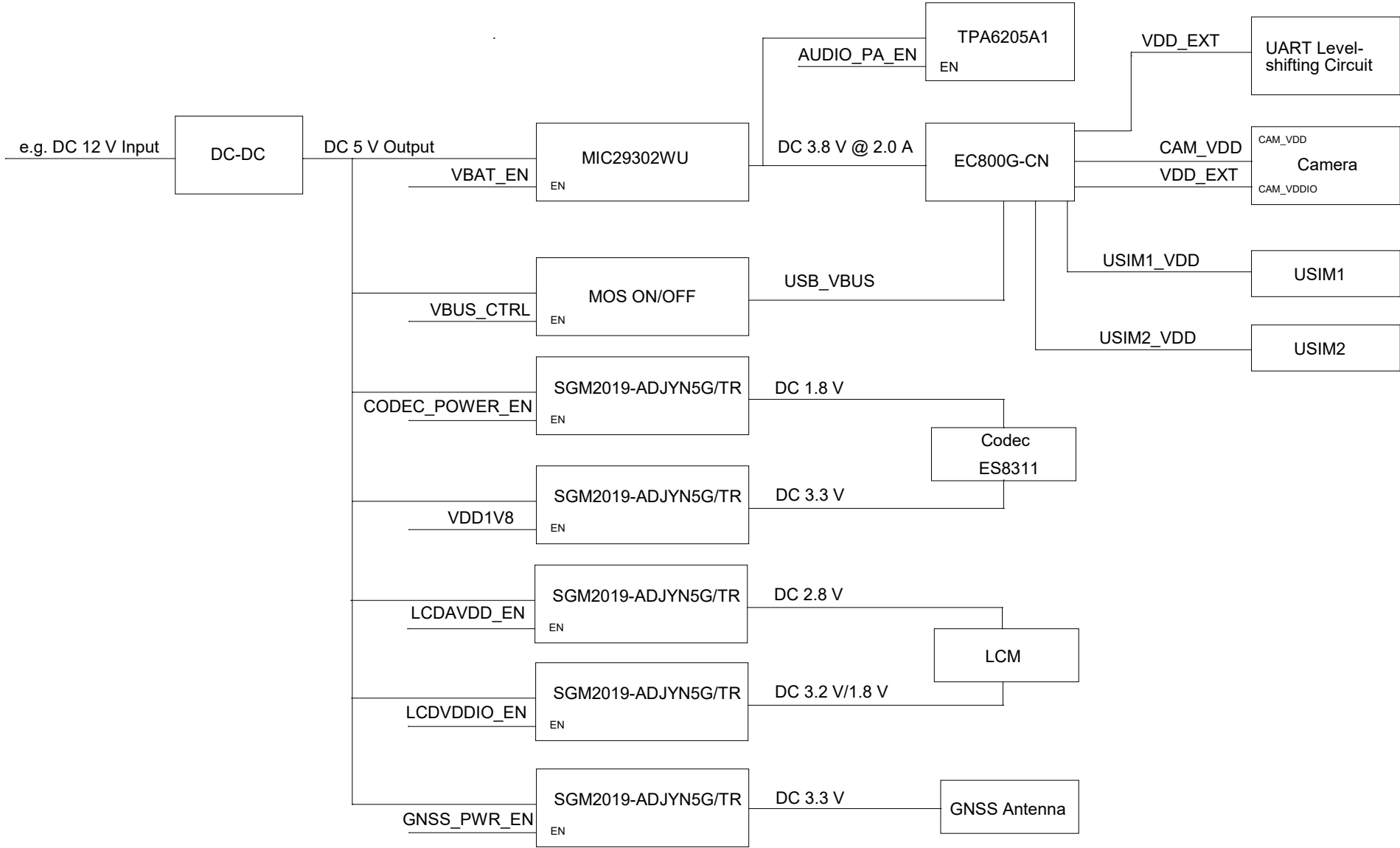
Block Diagram



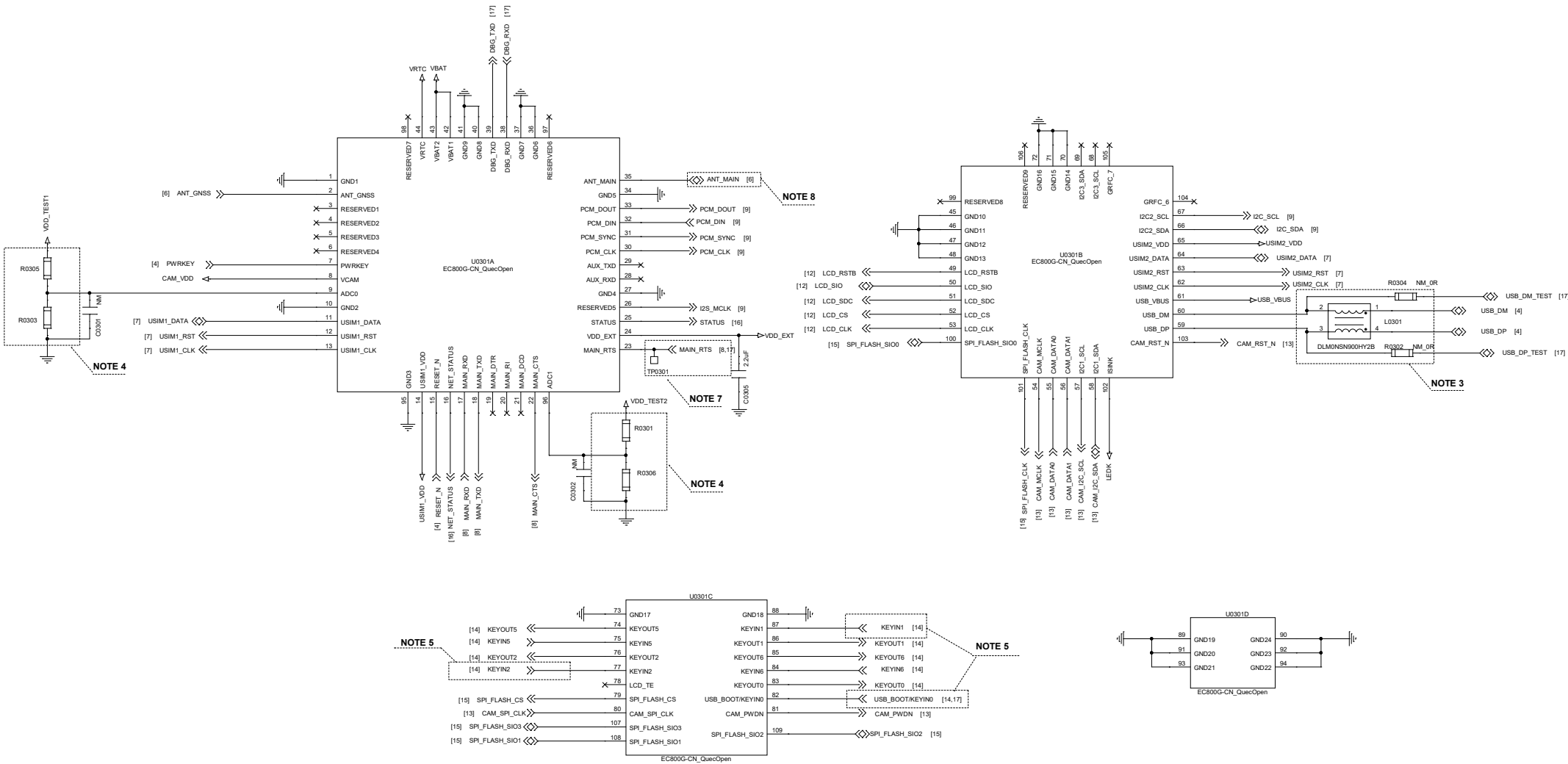
NOTE:

1. A transistor circuit or a voltage-level translator TXS0104EYZTR provided by Texas Instruments is recommended.
2. The power supply should be able to provide sufficient current of at least 2.0 A for the module.
3. The module supports Wi-Fi Scan function, which shares the same antenna interface with main antenna. The two functions cannot be used at the same time.
4. GNSS function is optional for the module.

Power System Block Diagram



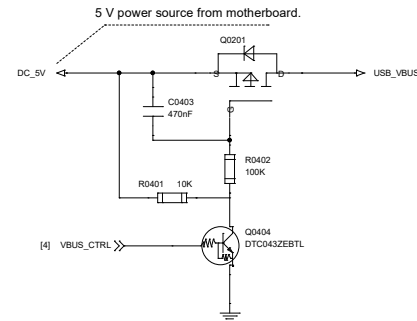
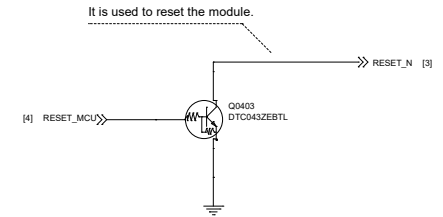
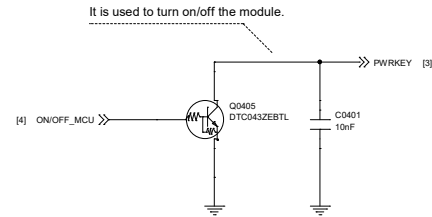
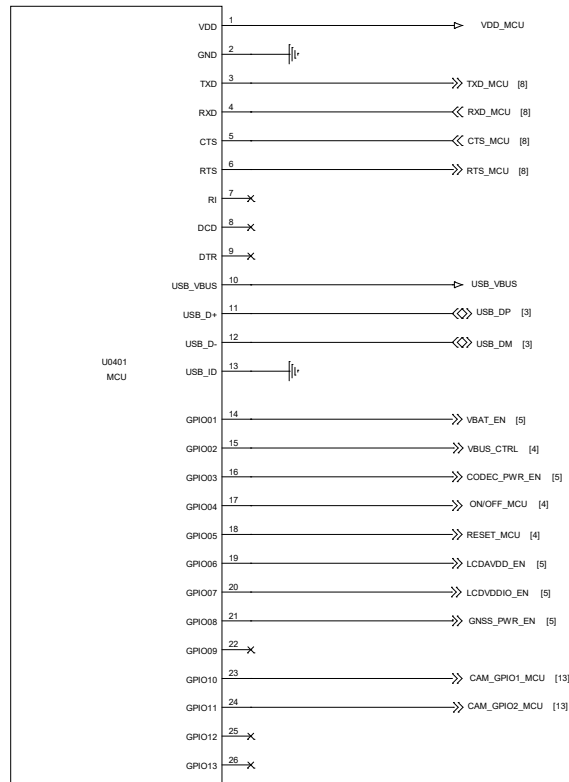
Module Interfaces



NOTE:

1. Keep unused and RESERVED pins unconnected, and all GND pins should be connected to the ground.
2. Ensure there is a complete reference ground plane below the module, and the ground plane is as close to the module layer as possible. It is recommended to design at least 4-layers.
3. A common mode choke L0301 is recommended to be added in series between the module and your MCU to suppress EMI spurious transmission.
Meanwhile, reserve the test points for upgrading the firmware over USB interface and minimize the extra stubs of the trace. L0301 and the two resistors R0302 and R0304 should be placed close to the module to ensure the integrity of USB signal.
4. Considering the difference of ADC voltage range among Quectel modules, when it is necessary to use ADC pins, it is strongly recommended to reserve the voltage divider circuit for better compatibility with other Quectel modules.
The resistance of the divider must be less than 100 k Ω , otherwise the measurement accuracy of the ADC will be significantly reduced. When the divider circuit is not used, the ADC pins require 1 k Ω resistors in series.
5. If emergency download is not required, USB_BOOT/KEYIN0 cannot be pulled down to low level or pulled up to high level before the module successfully starts up. KEYIN1, KEYIN2 cannot be pulled down to low level before the module successfully starts up.
6. MAIN_DTR, MAIN_RI and MAIN_DCD do not have the functions described by the pin names and their default functions are GPIOs.
7. A test point must be reserved for MAIN_RTS to capture CP logs.
8. The module supports Wi-Fi Scan function, which shares the same antenna interface with main antenna. The two functions cannot be used at the same time.

MCU Interfaces



NOTE:

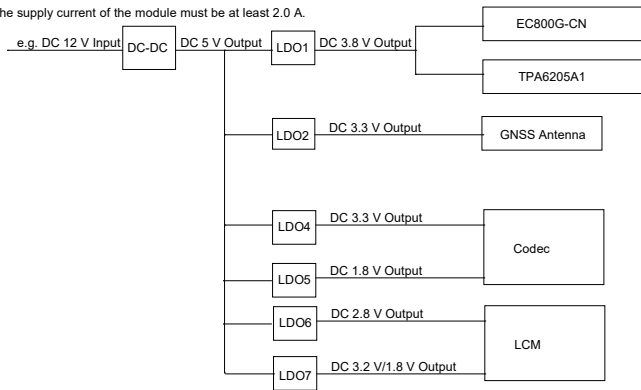
1. U0401 represents your MCU. The power domain of GPIO interfaces of the module is 1.8 V. If the GPIO interfaces of U0401 share the same power domain, the related level-shifting circuit can be omitted.
2. The USB 2.0 interface of the module only serves as a slave device and supports full-speed and high-speed modes. To communicate with the USB interface, MCU needs to support USB host mode or OTG function.
The USB_VBUS pin of the module should be powered by an external power system for USB detection, and VBUS_CTRL is used to turn on/off the USB_VBUS power supply.
3. It is recommended to select the default low-level GPIO pins of MCU as the control pins for PWRKEY and RESET_N of the module. Ensure that the maximum load capacitance of pins PWRKEY and RESET_N does not exceed 10 nF.

Power Supply Design

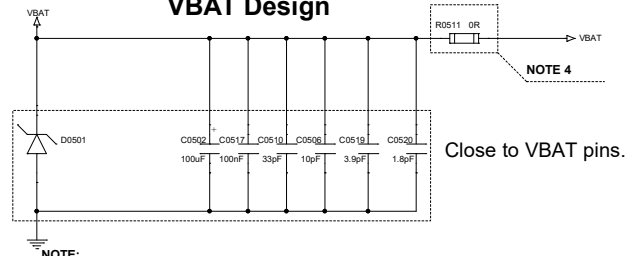
DC-DC Application

When the input voltage is above 7.0 V, use a DC-DC converter to convert the high input voltage into a 5.0 V first, and then use LDOs to convert it to 3.8 V, 3.3 V and 1.8 V to power the module, GNSS antenna, LCM, audio PA and codec.

The supply current of the module must be at least 2.0 A.



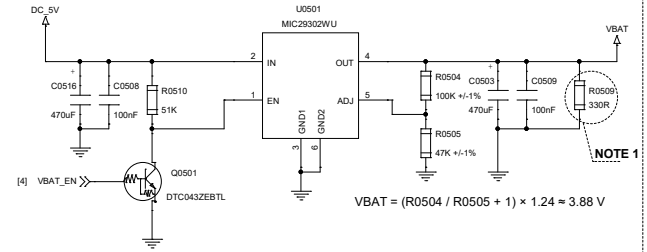
VBAT Design



NOTE:

1. The power supply should be able to provide sufficient current of at least 2.0 A for the module.
 2. The width of VBAT trace should not be less than 2 mm.
 3. The typical operating voltage of VBAT is 3.8 V.
 4. It is recommended to reserve a 0 Ω resistor (the size should not be smaller than 0603) for future debugging.
- The resistor should be placed close to VBAT pins.

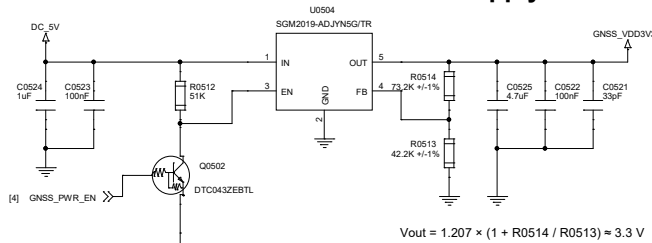
LDO Application



NOTE:

1. The recommended load current is greater than 10 mA.
2. When the input voltage is below 7.0 V, use an LDO to convert the input voltage to 3.8 V.

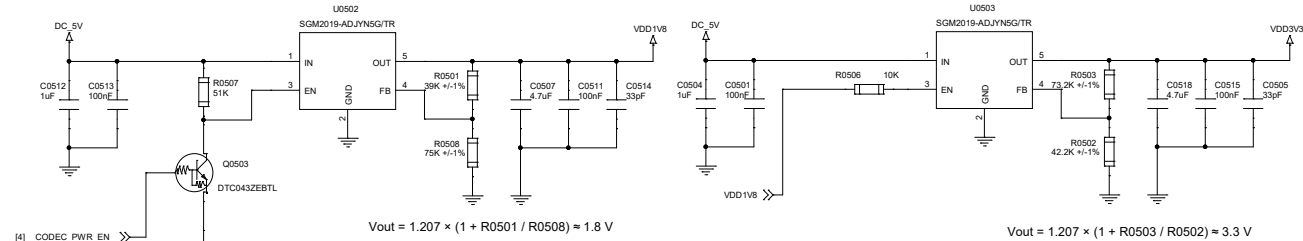
GNSS Antenna Power Supply



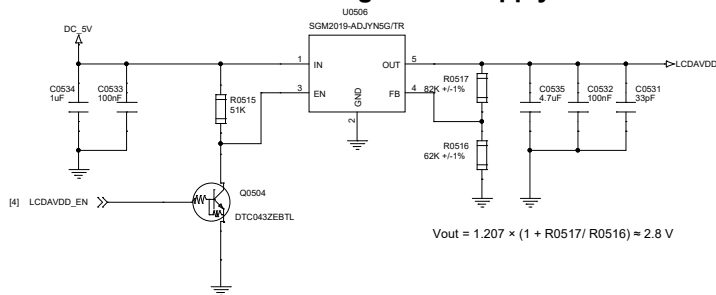
NOTE:

GNSS function of the module is optional.

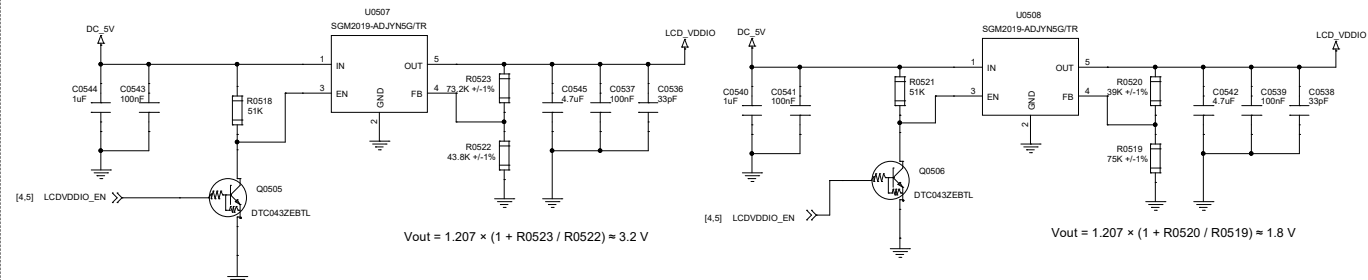
Codec Power Supply



LCM Analog Power Supply



LCM Digital Power Supply



NOTE:

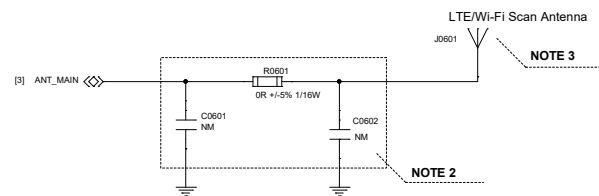
1. When the LCM is used in the 3.2 V voltage domain, refer to the design of the U0507.
2. When the LCM is used in the 1.8 V voltage domain, refer to the design of the U0508.

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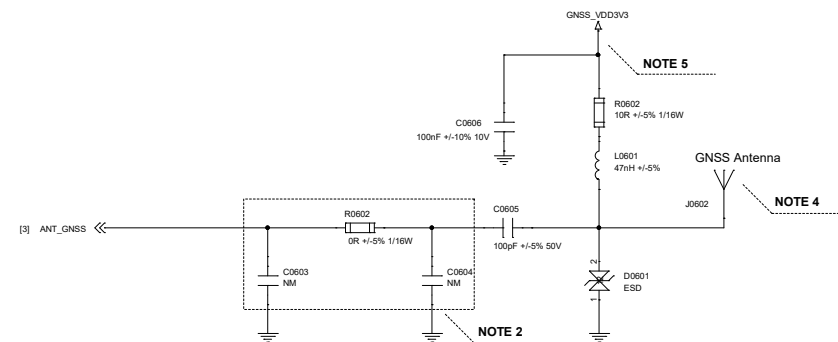
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DRWAN BY	Phoebe FUJiank WANG	CHECKED BY	Tiger CHENG
DATE	Wednesday, August 16, 2023	SIZE	A2
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Antenna Interface Design

LTE/Wi-Fi Scan Antenna Interface



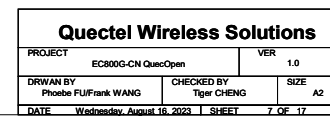
GNSS Antenna Interface



NOTE:

1. The single-ended impedance of the RF antenna is 50 Ω .
2. It is recommended to reserve a Π -type matching circuit at these antenna interfaces.
3. The module supports Wi-Fi Scan function, which shares the same antenna interface with main antenna. The two functions cannot be used at the same time.
4. GNSS function is optional for the module.
5. The active antenna uses an LDO for power supply, and GNSS_VDD3V3 circuit is not needed when using passive antenna.

USIM1

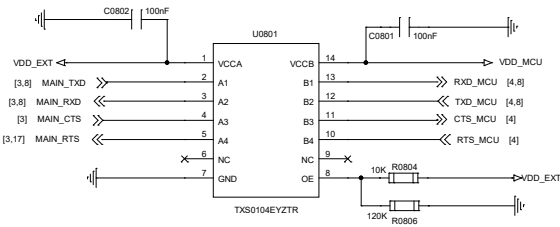


UART Interface Design

UART Level-shifting Circuit - Transistor Solution

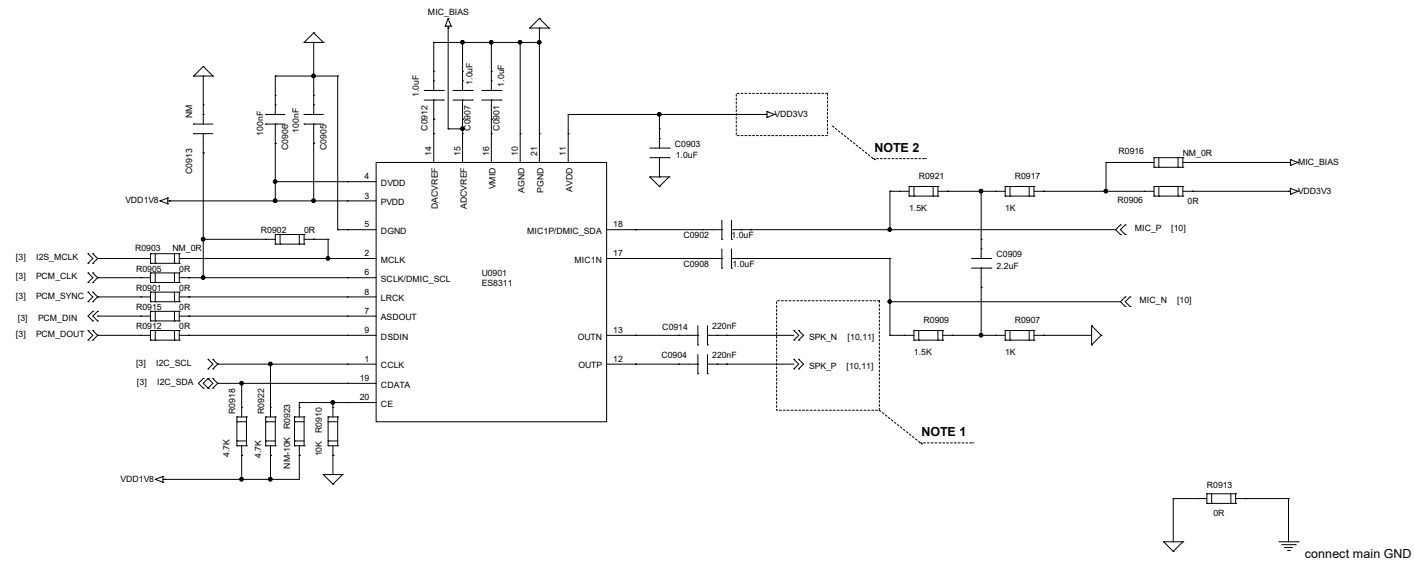


UART Level-shifting Circuit - IC Solution



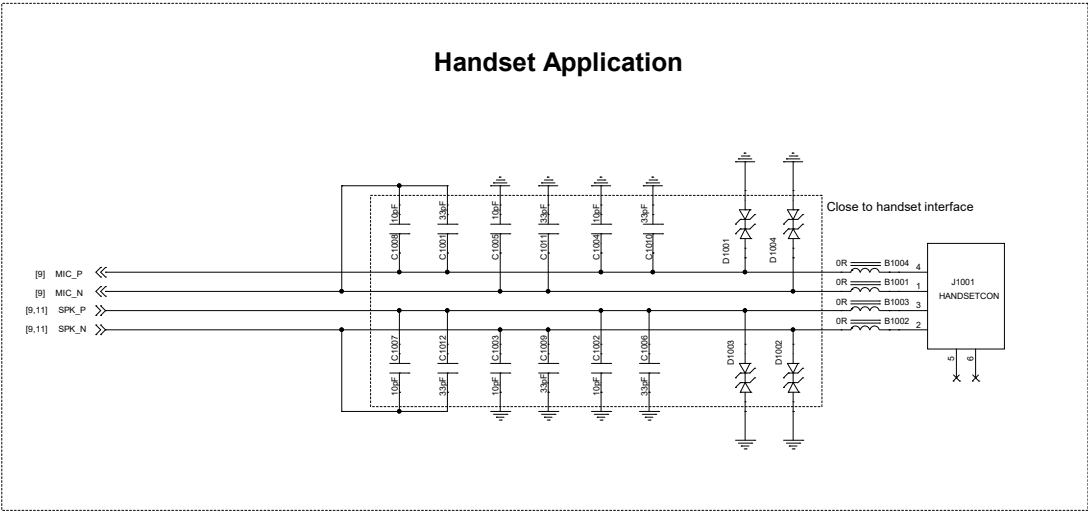
- NOTE:**
- There are two level-shifting solutions: transistor solution and IC solution, the latter of which is recommended.
 - The power supply of TXS0104EYZTR's VCCA should not exceed that of VCCB. For more information, refer to the datasheet of TXS0104EYZTR.
 - The transistor solution is not suitable for applications with baud rates exceeding 460 kbps. The capacitors C0803 and C0804 of 1 nF can improve the signal quality.
 - MAIN_RTS transistor circuit is similar to that of the MAIN_RXD. MAIN_CTS transistor circuit is similar to that of the MAIN_TXD.

Audio Codec Design (ES8311)



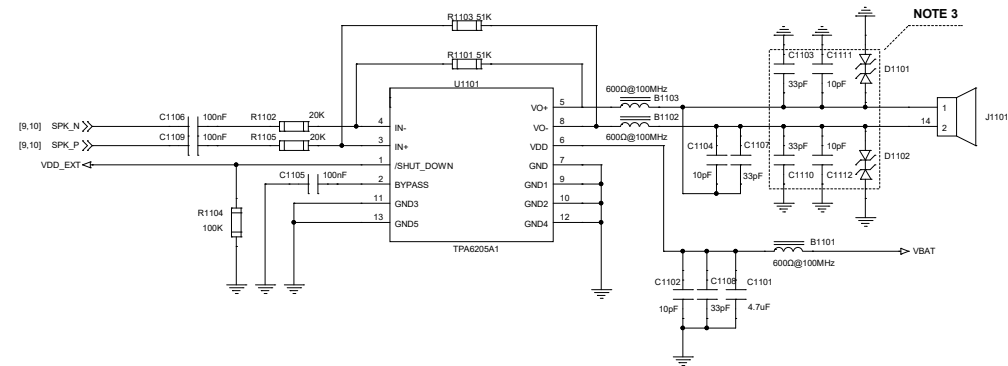
- NOTE:**
1. Differential signal that can be connected to audio power amplifier.
 2. AVDD requires a 3.3 V power supply by default.

Audio Codec Interface Design



- NOTE:**
- 1. The codec analog output can drive handset. For larger power loads such as loudspeaker, an audio power amplifier should be added in the design.
 - 2. In handset application, route the MIC and SPK signal traces as differential pairs respectively.
 - 3. All MIC and SPK signal traces shall be surrounded with ground on the layer and ground planes above and below, and far away from noises such as clock and DC-DC signals.

Analog Audio Design (Audio Power Amplifier)

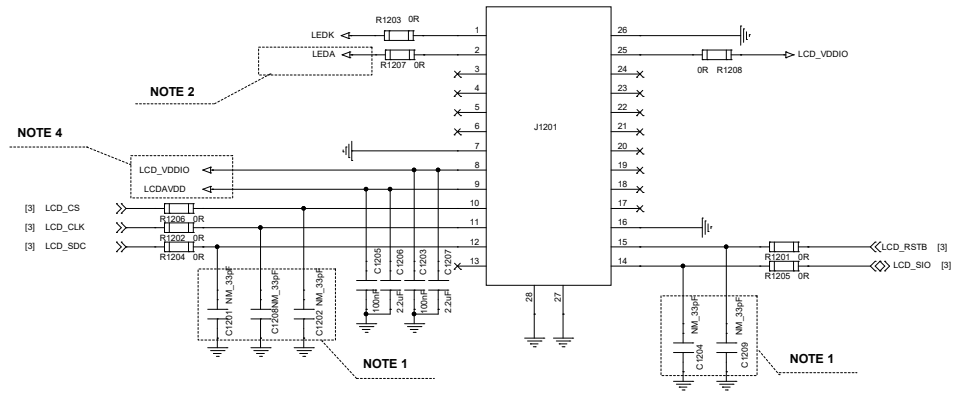


NOTE:

1. SPK_P and SPK_N are differential output channels that can be used for the external audio amplifier.
When channels are turned on or off, it is possible for the module to emit a pop sound, which can be eliminated by controlling the enable pin of the audio PA. For details, contact Quectel Technical Support.
2. The amplifier model in this design is for reference only. Select the audio power amplifier with appropriate power according to actual needs.
3. Place filter capacitors and ESD protection components close to the loudspeaker.
4. The selection of ESD protection components is related to the selection of audio power amplifier. Ensure that the output voltage of the audio power amplifier is within the maximum reverse working voltage range of ESD protection components under normal working condition, so as to avoid damage to the components.

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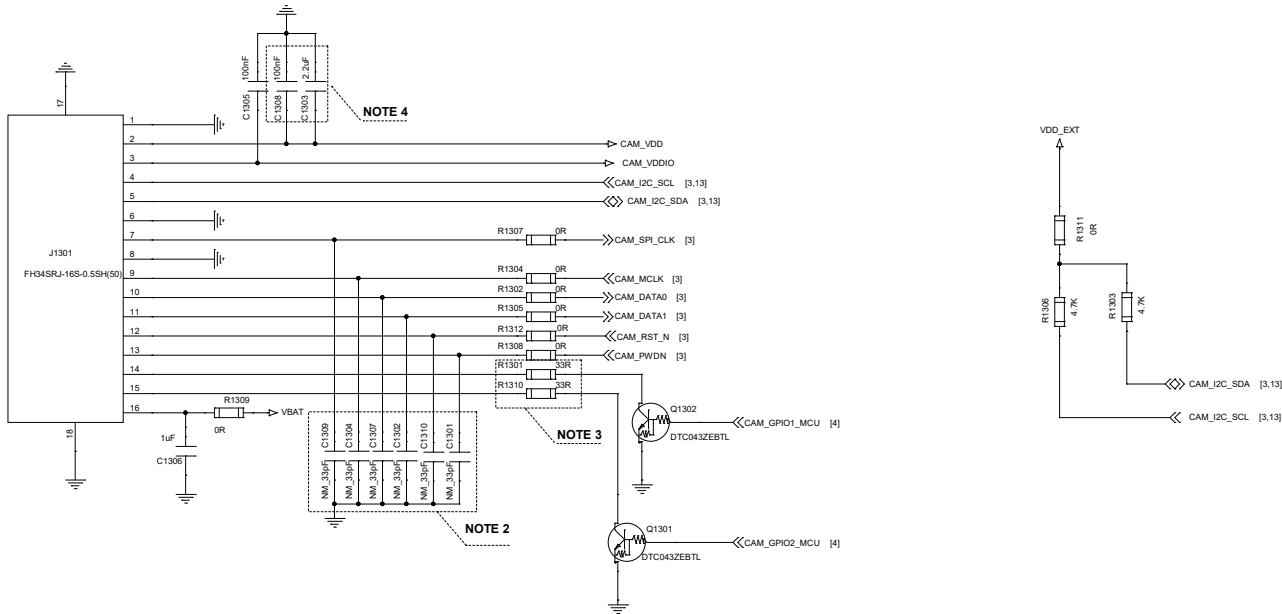
LCM Interface Design



NOTE:

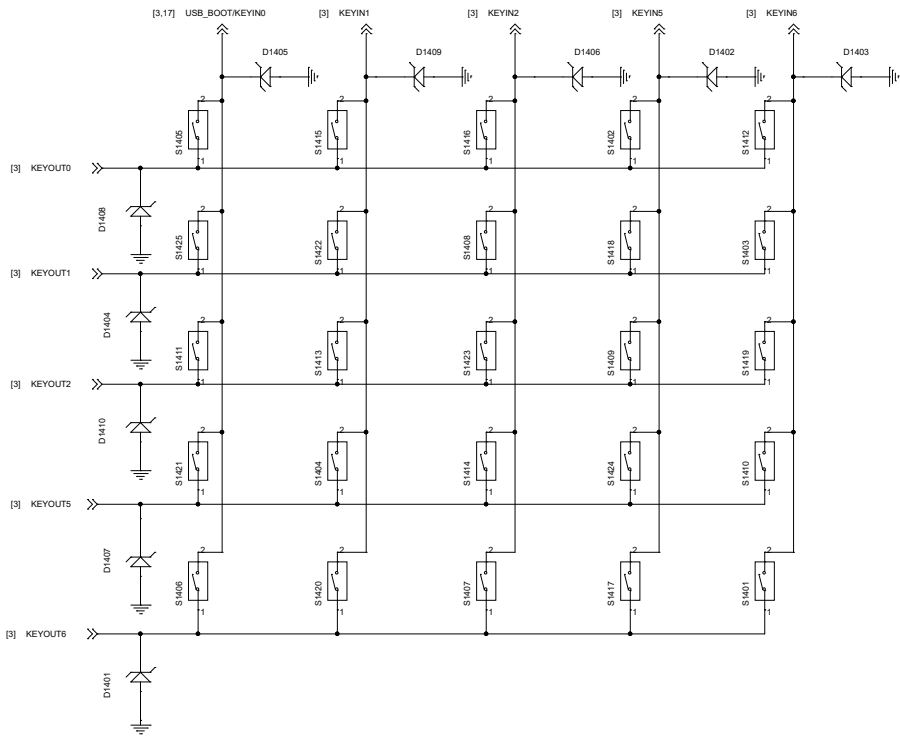
1. The 33 pF capacitors of the signal pins are reserved, and can be used as per the actual debugging situation.
2. The power supply pin LEDA of the backlight is provided by external power supply circuit, and you can design the circuit by yourself.
3. It is strongly recommended to use switch control for all power supplies, otherwise there may be current leakage and power consumption increase.
4. In order to avoid abnormal LCD display caused by power fluctuation, the filter capacitors of the LCD power supply pins, LCDAVDD and LCD_VDDIO, must be mounted.

Camera Interface Design



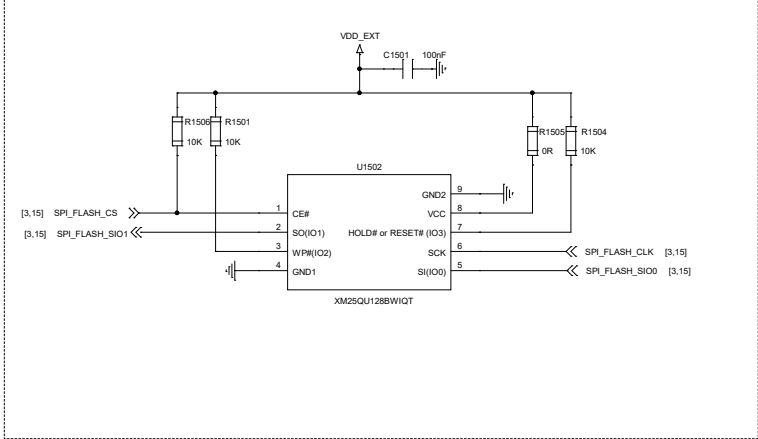
- NOTE:**
1. By controlling the triode switching circuit, CAM_GPIO1_MCU controls the cathode of the positioning light of the camera, and CAM_GPIO2_MCU controls the cathode of the supplement light of the camera. It is recommended to select default pull-down GPIO pins as the two control pins.
 2. The 33 pF capacitors of signal pins should be reserved, and be used according to actual debugging situation.
 3. The values of current limiting resistors of positioning light and supplement light (R1301 and R1310) should be varied according to the required brightness.
 4. The capacitors of the CAM_VDD power supply should be connected to the GND layer directly, otherwise, power supply noise may lead to abnormalities such as white dots on the preview screen.
 5. It is recommended to design camera power supply by yourself.
 6. It is strongly recommended to use switch control for all power supplies, otherwise there may be current leakage and power consumption increase.

Matrix Keypad Interface Design

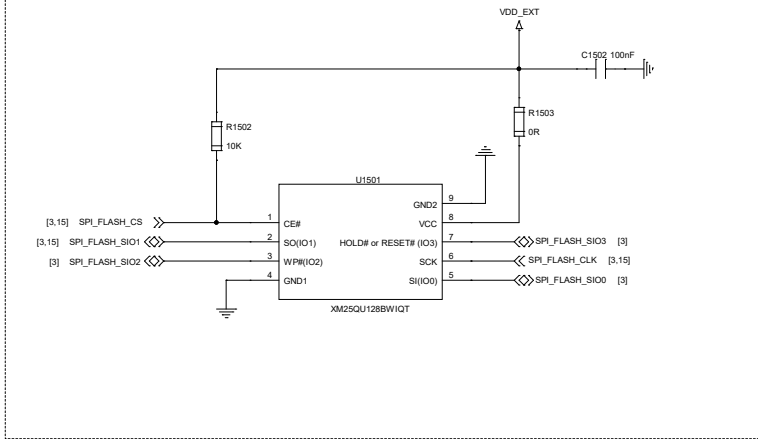


External Flash Interface Design

Four-wire NOR Flash Circuit

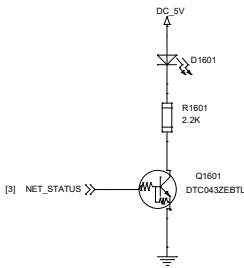
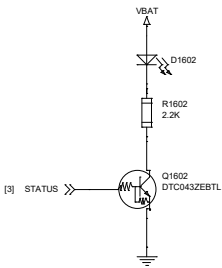


Six-wire NOR Flash Circuit



- NOTE:**
1. The filter capacitors C1501 and C1502 need to be placed close to the power supply pins of flash chip.
 2. For the four-wire NOR flash circuit, it is recommended to connect WP and HOLD pins with pull-up resistors to avoid damaging flash chips caused by abnormal levels, resulting in abnormal transmission or data loss.

Indicators

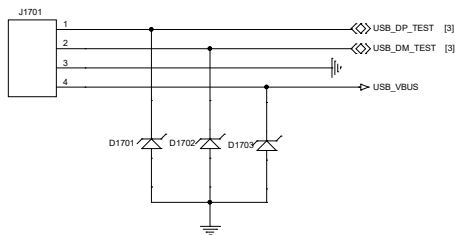


NOTE:

- For more details about STATUS and NET_STATUS, see the hardware design document of the module.
- If the low current consumption is required when your application is in sleep status, replace the power supplies (VBAT and DC_5V) of STATUS and NET_STATUS indication LEDs with the external controllable ones, which can be turned off to reduce the power consumption during module sleep status.

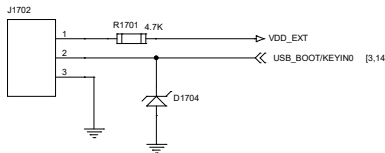
Reserved Test Points

USB Interface Test Points



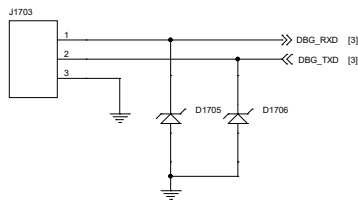
- NOTE:**
1. Test points must be reserved for USB interface.
 2. The module can upgrade firmware and debug the software over USB interface. The parasitic capacitance of the ESD protection components on USB data traces should not exceed 2 pF.

Emergency Download Interface Test Points



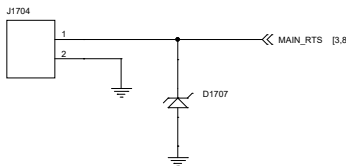
- NOTE:**
1. It is recommended to reserve a test point for USB_BOOT/KEYIN0 interface.
 2. Pull down the USB_BOOT/KEYIN0 to GND before the module starts up, and the module will enter the emergency download mode when it is turned on. In this mode, the module supports firmware upgrading over USB interface.

Debug UART Test Points



- NOTE:**
1. Test points must be reserved for DBG_TXD and DBG_RXD to capture AP logs.
 2. The debug UART interface supports 1.8 V power domain, and a level-shifting circuit should be used if the power domain of your application is 3.3 V. For details, refer to the "UART Interface Design".
 3. The debug UART only supports the baud rate of 2000000 bps.

MAIN_RTS Test Point



- NOTE:**
1. A test point must be reserved for MAIN_RTS to capture CP logs.
 2. MAIN_RTS supports 1.8 V power domain, and a level-shifting circuit should be used if the power domain of your application is 3.3 V. For details, refer to the "UART Interface Design".
 3. When capturing the CP log, set the baud rate to 8000000 bps.